作业（4月1日）

pp81-82 习题8-2

7, 8, 11, 12, 13, 15, 16, 17, 18

[注意：如果 $z=f(x,y)$ 在区域D内的偏导函数都存在，则函数 $z=f(x,y)$ 的全微分为 $dz=\frac{\partial z}{\partial x}dx+\frac{\partial z}{\partial y}dy$]

作业（4月3日）

pp89-90

2, 4, 8, 9, 10, 15

补充题（由p86 例5改编而成）

设变换 $\begin{cases} u=x+by \\ v=x+ay \end{cases}$ 可把方程 $6\frac{\partial^2 z}{\partial x^2}+\frac{\partial^2 z}{\partial x\partial y}-\frac{\partial^2 z}{\partial y^2}=0$ 简化为 $\frac{\partial^2 z}{\partial u\partial v}=0$，且 z 具有连续的二阶偏导数，求常数 a,b.

15. 引入新的自变量 $\xi=x-at,\ \eta=x+at$ 化简方程 $\frac{\partial^2 u}{\partial t^2}=a^2\frac{\partial^2 u}{\partial x^2}$ [波动方程/双曲方程]

$\frac{\partial^2 u}{\partial t^2}-a^2\Delta u=f$

解：[解PDE 默认充分光滑]

$\frac{\partial u}{\partial t}=\frac{\partial u}{\partial \xi}\cdot\frac{\partial \xi}{\partial t}+\frac{\partial u}{\partial \eta}\cdot\frac{\partial \eta}{\partial t}=-a\frac{\partial u}{\partial \xi}+a\frac{\partial u}{\partial \eta}$ $\qquad$ $\frac{\partial u}{\partial x}=\frac{\partial u}{\partial \xi}+\frac{\partial u}{\partial \eta}$

$\frac{\partial^2 u}{\partial t^2}=\frac{\partial}{\partial t}\left(\frac{\partial u}{\partial t}\right)=\boxed{\left[\frac{\partial}{\partial \xi}\cdot\frac{\partial \xi}{\partial t}+\frac{\partial}{\partial \eta}\cdot\frac{\partial \eta}{\partial t}\right]}\left(\frac{\partial u}{\partial t}\right)=\left[-a\frac{\partial}{\partial \xi}+a\frac{\partial}{\partial \eta}\right]\left(-a\frac{\partial u}{\partial \xi}+a\frac{\partial u}{\partial \eta}\right)$ 高阶偏导化偏向对算子运算

$=a^2\left(\frac{\partial^2 u}{\partial \xi^2}-2\underline{\frac{\partial^2 u}{\partial \xi\partial \eta}}+\frac{\partial^2 u}{\partial \eta^2}\right)$ ① 假定 u 充分光滑，$\frac{\partial^2 u}{\partial \xi\partial \eta}=\frac{\partial^2 u}{\partial \eta\partial \xi}$

$\frac{\partial^2 u}{\partial x^2}=\frac{\partial^2 u}{\partial \xi^2}+2\frac{\partial^2 u}{\partial \xi\partial \eta}+\frac{\partial^2 u}{\partial \eta^2}$ ②

① ②代回 $u_{tt}=a^2u_{xx}$ $\Rightarrow$ $a^2\frac{\partial^2 u}{\partial \xi\partial \eta}=0$ a是不为0的常数 $u_{\xi\eta}=0$

$u=u(\xi,\eta)$ 无交叉偏导，说明 $u(\xi,\eta)$ 无形如 $f(\xi)\cdot g(\xi)$ 这样的项，也就是 ξ η 可分离

(例如 $\xi^\alpha\eta^\beta$)

$\frac{\partial^2 u}{\partial \xi\partial \eta}=0$ $\Rightarrow$ $\frac{\partial u}{\partial \xi}=f(\xi)$ $\qquad$ $u=\int f(\xi)d\xi+\varphi_2(\eta)=\varphi_1(\xi)+\varphi_2(\eta)$

[不做要求] 为什么引入 $\xi=x-at$ $\eta=x+at$ 就能简化?

定义算子"□"：$\square u=\frac{\partial^2 u}{\partial t^2}-a^2\frac{\partial^2 u}{\partial x^2}=0$ $\qquad$ $\square u=\underbrace{\left[\frac{\partial}{\partial t}-a\frac{\partial}{\partial x}\right]\left[\frac{\partial}{\partial t}+a\frac{\partial}{\partial x}\right]}_{\square}u$

$\begin{cases} \underline{\frac{\partial}{\partial t}u+a\frac{\partial}{\partial x}u}=v \\ \frac{\partial}{\partial t}v-a\frac{\partial}{\partial x}v=0 \end{cases}$

$\frac{\partial u}{\partial t}+\frac{\partial u}{\partial x}\cdot a$ $\qquad$ $\frac{\partial u}{\partial t}+\frac{\partial u}{\partial x}\cdot\boxed{\frac{\partial x}{\partial t}}$ $\qquad$ $\frac{dx}{dt}=a$ $\qquad$ $x=at$ $\qquad$ $\xi=x-at$

$\frac{\partial u}{\partial t}+\frac{\partial u}{\partial x}[-a]$ $\qquad$ $x=-at$ $\qquad$ $\eta=x+at$

补充题：和T15一样 $\frac{\partial z}{\partial x}=\frac{\partial z}{\partial u}\cdot\frac{\partial u}{\partial x}+\frac{\partial z}{\partial v}\cdot\frac{\partial v}{\partial x}$

$\frac{\partial^2 z}{\partial x^2}=\frac{\partial^2 z}{\partial u^2}+2\frac{\partial^2 z}{\partial u\partial v}+\frac{\partial^2 z}{\partial v^2}=0$

同理 $\frac{\partial^2 z}{\partial x\partial y}$，$\frac{\partial^2 z}{\partial y^2}$，用 $u\ v$ 偏导表示

代回原方程 $6z''_{xx}+z''_{xy}-z''_{yy}=0$

$\Rightarrow (6+b-b^2)z''_{uu}+(12+a+b-2ab)z''_{uv}+(6+a-a^2)z''_{vv}=0$

与结果 $z''_{uv}=0$ 比较

$$\begin{cases}6+b-b^2=0 & ①\\ 6+a-a^2=0 & ②\\ 12+a+b-2ab\neq 0 & ③\end{cases}$$

3，-2 ✕ 3，-2

$a\neq b$

$\begin{cases}a=3\\ b=-2\end{cases}$ $\begin{cases}a=-2\\ b=3\end{cases}$

↳ ③-①-② $\Rightarrow (a-b)^2\neq 0$

习题8-2，T7. $\frac{\partial^3 u}{\partial x\partial y\partial z}\neq\frac{\partial u}{\partial x}\cdot\frac{\partial u}{\partial y}\cdot\frac{\partial u}{\partial z}$

算子复合：$\frac{\partial^3}{\partial x\partial y\partial z}u=\frac{\partial}{\partial y}\circ\frac{\partial}{\partial y}\circ\frac{\partial}{\partial z}u$

习题8-3，T4. $\frac{\partial^2 u}{\partial y^2}=\frac{2x}{y^3}f'_2+\frac{x^2}{y^4}f''_{22}$

y^3，不是 y

PS. 期中考：T13. $\boxed{\text{经典伪证}}$

$\lim\limits_{n\to\infty}\frac{\ln a_n}{\ln n}=1 \not\Rightarrow a_n\sim n$ 不要随便把函数扔了！

反例：取 $a_n=n\cdot\ln n$ $\frac{\ln a_n}{\ln n}=\frac{\ln n+\ln\ln n}{\ln n}=\frac{t+\ln t}{t}\to 1$ $t=\ln n\to+\infty$

微积分习题课材料

1. 设 n 为整数，若 $\forall t>0$，$f(tx,ty)=t^n f(x,y)$，则称 f 是 n 次齐次函数。特别地，若 $\forall t>0$，$f(tx,ty)=f(x,y)$，则称 f 是零次齐次函数。试证明，$f(x,y)$ 是零次齐次函数 $\Leftrightarrow x\dfrac{\partial f}{\partial x}+y\dfrac{\partial f}{\partial y}=0$。

2. 设 $\begin{cases}\xi=2xy\\ \eta=x^2-y^2\end{cases}(xy\neq 0)$，解下列方程（解可含任意函数）：

(1) $y\dfrac{\partial u}{\partial x}+x\dfrac{\partial u}{\partial y}=0$;

(2) $x\dfrac{\partial u}{\partial x}-y\dfrac{\partial u}{\partial y}=0$。

3 2. 设 z 具有连续的二阶偏导数，问是否存在常数 a,b，使在变换 $\begin{cases}u=\dfrac{y}{x}\\ v=ax+by\end{cases}$ 下，将方程 $x^2z''_{xx}+2xyz''_{xy}+y^2z''_{yy}=0$ 变为 $z''_{vv}=0$？

4 6. 设 $w=3x^2+2y^2+z^2$，其中 $z=f(x,y)$ 是由方程 $x^2+y^2+z^2-2x+2y-4z-10=0$ 所确定的隐函数，求 w_x 及 w_{xx}。

5 7.（08 - 09 真题）设 $z=f(x,y)$，$x=g(y,z)$ 均可微，求 $\dfrac{\partial z}{\partial x}$（设解答中出现的分母不为零）。

6 取 x 作为函数，而 $u=y-z$，$v=y+z$ 作为自变量，变换方程 $(y-z)\dfrac{\partial z}{\partial x}+(y+z)\dfrac{\partial z}{\partial y}=0$。

7 6. 设函数 $z=z(x,y)$ 满足方程 $x\dfrac{\partial^2 z}{\partial x^2}+2\dfrac{\partial z}{\partial x}=\dfrac{y}{x}$，试以 $\xi=\dfrac{y}{x}$，$\eta=y$ 为新的自变量，$u=yz-x$ 为 ξ,η 的函数，把上述方程变换为 $u=u(\xi,\eta)$ 所满足的方程（本题参考答案为 $\dfrac{\partial^2 u}{\partial \xi^2}=0$）。

1 ⇒：

$f(xt,yt)=f(x,y)$，两边对 t 求导

$\frac{\partial f}{\partial x}\cdot x+\frac{\partial f}{\partial y}\cdot y=0$

⇐： x,y 看成 t 的参数

令 $g(t)=f(xt,yt)$　　$g'(t)=\frac{\partial f}{\partial x}\cdot x+\frac{\partial f}{\partial y}\cdot y=0$

$g(t)\equiv C\equiv g(1)$，即 $f(tx,ty)=f(x,y)$

2 $\dfrac{\partial(\xi,\eta)}{\partial(x,y)}=\begin{vmatrix}\frac{\partial\xi}{\partial x} & \frac{\partial\xi}{\partial y}\\ \frac{\partial\eta}{\partial x} & \frac{\partial\eta}{\partial y}\end{vmatrix}=\begin{vmatrix}2y & 2x\\ 2x & -2y\end{vmatrix}=-4(x^2+y^2)$

可逆

$\frac{\partial u}{\partial x}=\frac{\partial u}{\partial \xi}\cdot\frac{\partial \xi}{\partial x}+\frac{\partial u}{\partial \eta}\cdot\frac{\partial \eta}{\partial x}=2y\frac{\partial u}{\partial \xi}+2x\cdot\frac{\partial u}{\partial \eta}$

$\frac{\partial u}{\partial y}=2x\frac{\partial u}{\partial \xi}+(-2y)\frac{\partial u}{\partial \eta}$

代入① $(x^2+y^2)\frac{\partial u}{\partial \xi}=0$ ⇒ $\frac{\partial u}{\partial \xi}=0$　　$u=\varphi_1(\eta)=\varphi_1(x^2-y^2)$

代入② $(x^2+y^2)\frac{\partial u}{\partial \eta}=0$ ⇒ $\frac{\partial u}{\partial \eta}=0$　　$u=\varphi_2(\xi)=\varphi_2(2xy)$

3 $\frac{\partial}{\partial x}=[\frac{\partial}{\partial u}\cdot\frac{\partial u}{\partial x}+\frac{\partial}{\partial v}\cdot\frac{\partial v}{\partial x}]=[-\frac{y}{x^2}\frac{\partial}{\partial u}+a\frac{\partial}{\partial v}]$

$\frac{\partial}{\partial y}=[\frac{\partial}{\partial u}\cdot\frac{\partial u}{\partial y}+\frac{\partial}{\partial v}\cdot\frac{\partial v}{\partial y}]=[\frac{1}{x}\frac{\partial}{\partial u}+b\frac{\partial}{\partial v}]$

$z''_{xx}=\dfrac{a^2x^4\frac{\partial^2 z}{\partial v^2}-2ax^2y\cdot\frac{\partial^2 z}{\partial u\partial v}+b^2\frac{\partial^2 z}{\partial v^2}}{x^4}$，$z''_{xy}$，$z''_{yy}$ ...

代入原方程 $(a^2x^2+2abxy+b^2y^2)z''_{vv}=0$　　取 $a=1$，$b=0$ 即可

4 $x^2+y^2+z^2-2x+2y-4z-10=0$　两边对 x 求导：$2x+2z\cdot z_x-2-4z_x=0$ ⇒ $z_x=-\frac{x-1}{z-2}$　$(z\neq 2)$

$w_x=6x+2z\cdot z_x=6x-2z\cdot\frac{x-1}{z-2}$　　$w_{xx}=6+2z_x^2+2zz_{xx}=\ldots$

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5. $\boxed{\text{直接求}}$

$$\frac{\partial z}{\partial x}=\frac{\partial f}{\partial x}+\frac{\partial f}{\partial y}\cdot\frac{\partial y}{\partial x}\quad ①$$

$$\frac{\partial x}{\partial y}=\frac{\partial g}{\partial y}+\frac{\partial g}{\partial z}\cdot\frac{\partial z}{\partial y}\quad ②$$

$$\frac{\partial z}{\partial y}=\frac{\partial f}{\partial x}\cdot\frac{\partial x}{\partial y}+\frac{\partial f}{\partial y}\quad ③$$

由②③ $\dfrac{\partial x}{\partial y}=g_y+g_z\left(f_x\cdot\dfrac{\partial x}{\partial y}+f_y\right)$

$$\frac{\partial x}{\partial y}=\frac{g_y+g_zf_y}{1-g_zf_x}\ \Rightarrow\ \frac{\partial y}{\partial x}=\frac{1-g_zf_x}{g_y+g_zf_y}$$

则 $\dfrac{\partial z}{\partial x}=f_x+f_y\dfrac{\partial y}{\partial x}=\dfrac{f_xg_y+f_y}{g_y+g_zf_y}$

注：$x=g(y,z)=g(y,f(x,y))\Rightarrow$ 可写成隐函数 $F(x,y)=0$ 的形式

$F_xdx+F_ydy=0$

则 $\dfrac{\partial y}{\partial x}=-\dfrac{F_x}{F_y}$ $\quad\dfrac{\partial x}{\partial y}=-\dfrac{F_y}{F_x}$ $\quad\Rightarrow\dfrac{\partial y}{\partial x}\cdot\dfrac{\partial x}{\partial y}=1$

直观理解 隐函数 $F(x,y)=0$：x 变化"1"，y 变化"△"，则 y 变化"1"时，x 变化"$\frac{1}{\triangle}$"

☆ 但 $F(x,y,z)=0$：$\dfrac{\partial x}{\partial y}\cdot\dfrac{\partial y}{\partial z}\cdot\dfrac{\partial z}{\partial x}$ 不一定为1

另法：构造 $\boxed{\text{隐函数}}$

$$\begin{cases}F(x,y,z)=f(x,y)-z=0\\ G(x,y,z)=g(y,z)-x=0\end{cases}\xrightarrow{F,G\text{对}x\text{偏导}}\begin{cases}\dfrac{\partial f}{\partial x}+\dfrac{\partial f}{\partial y}\cdot\dfrac{\partial y}{\partial x}-\dfrac{\partial z}{\partial x}=0\\ \dfrac{\partial g}{\partial y}\cdot\dfrac{\partial y}{\partial x}+\dfrac{\partial g}{\partial z}\cdot\dfrac{\partial z}{\partial x}-1=0\end{cases}$$

解方程组 $\begin{cases}f_y\cdot\dfrac{\partial y}{\partial x}-\dfrac{\partial z}{\partial x}=-f_x\\ g_y\cdot\dfrac{\partial y}{\partial x}+g_z\cdot\dfrac{\partial z}{\partial x}=1\end{cases}$

6，7 本质是一样的

不太好找链式法则就去构造微分

6. $du=d(y-z)=dy-dz\qquad dv=dy+dz$

[想办法构造 $dz=\underbrace{f_1(x_u,x_v)}_{}^{\frac{\partial z}{\partial x}}dx+\underbrace{f_2(x_u,x_v)}_{}^{\frac{\partial z}{\partial y}}dy$]

$$dx=dx(u,v)=\frac{\partial x}{\partial u}du+\frac{\partial x}{\partial v}dv=\frac{\partial x}{\partial u}(dy-dz)+\frac{\partial x}{\partial v}(dy+dz)$$

$$=(x_u+x_v)dy+(-x_u+x_v)dz$$

$\dfrac{\partial z}{\partial x}=\dfrac{1}{-x_u+x_v}\triangleq f_1(x_u,x_v)\qquad\dfrac{\partial z}{\partial y}=-\dfrac{x_u+x_v}{-x_u+x_v}\triangleq f_2(x_u,x_v)$

代回原式 化简得 $\quad u-v\cdot(x_u+x_v)=0\quad\Rightarrow x_u+x_v=\dfrac{u}{v}$

7. 条件 $\xi = \frac{y}{x}$, $\eta = y \Leftrightarrow x = \frac{\eta}{\xi}$, $y = \eta$

$$\begin{cases} d\xi = -\frac{y}{x^2}dx + \frac{1}{x}dy \\ d\eta = dy \end{cases} \Leftrightarrow \begin{cases} dx = -\frac{\eta}{\xi^2}d\xi + \frac{1}{\xi}d\eta \\ dy = d\eta \end{cases}$$

要构造 $dz = \frac{\partial z}{\partial x}dx + \frac{\partial z}{\partial y}dy$

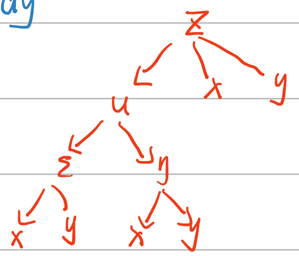
$u = yz - x$ $\quad du = d(yz - x) = -dx + z\,dy + y\,dz$ ①

另一方面 $u = u(\xi, \eta)$ $\quad du = \frac{\partial u}{\partial \xi}d\xi + \frac{\partial u}{\partial \eta}d\eta = u_\xi \cdot [-\frac{y}{x^2}dx + \frac{1}{x}dy] + u_\eta\,dy$ ②

①,② 联立 $\quad y\,dz = (1 - \frac{y u_\xi}{x^2})dx + (\frac{u_\xi}{x} + u_\eta - z)dy$

得到 $\frac{\partial z}{\partial x} = \frac{1}{y} - \frac{1}{x^2}u_\xi$

实际链子



则 $\frac{\partial^2 z}{\partial x^2} = \frac{\partial}{\partial x}[\frac{1}{y} - \frac{1}{x^2}u_\xi] = \frac{2}{x^3}u_\xi + \frac{1}{x^2}\frac{\partial}{\partial x}u_\xi$

$\frac{\partial}{\partial x}u_\xi = \frac{\partial}{\partial \xi}u_\xi \cdot \frac{\partial \xi}{\partial x} + \frac{\partial}{\partial \eta}u_\xi \cdot \frac{\partial \eta}{\partial x} = u_{\xi\xi} \cdot (-\frac{y}{x^2})$

代入方程 [题目好像出错了，得不到 $u_{\xi\xi} = 0$]